MAJ Research Progress Report

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1. Overview

This report summarizes the full progress of the Memory-Augmented Judge (MAJ) research project. The core hypothesis is that augmenting an LLM-based judge with a structured memory system storing past evaluation experiences leads to more accurate, consistent, and generalizable judgments compared to a stateless judge.

2. System Architecture MAJ Core

The MAJ system is built around a **3-stage memory-augmented retrieval pipeline**:

Stage	Description
Stage 1	**Similar Policy Retrieval** finds past evaluations with similar policies or rules
Stage 2	**Contrasting Examples** pulls good vs. bad examples to give the judge a clear reference signal
Stage 3	**Semantic Pattern Matching** retrieves semantic patterns from past issues, classified on the fly by the LLM

Key Implementation Details

- Semantic nodes automatically classify issues into categories using on-the-fly LLM classification
- Memory is stored as a graph structure, enabling structured retrieval across all three stages
- Similarity thresholds were tuned to prevent over-flagging of safe code (false positives)
- Memory can be initialized organically (learned during evaluation) or seeded with ground truth examples

3. Experiments & Results

Experiment 1 Unit Test Evaluation (Code Evaluation)

Setup: Evaluated MAJ on a set of unit test cases for code evaluation tasks.

Setting	Accuracy
Stateless (no memory)	90%
With Memory	100%
Improvement	**+10%**

Key Insight: Memory provides a perfect accuracy boost on structured code evaluation tasks where past policies are directly reusable.

Experiment 2 EvalsBench Model Comparison (100 & 1000 Samples)

Setup: Tested MAJ across three models of varying reasoning capability on the EvalsBench dataset. Experiments were run at both 100 and 1000 samples to validate consistency at scale.

Model	Stateless	With Memory	Gain
GPT-4o-mini (weak)	60%	60%	0% (neutral)
GPT-4o	78%	85%	+7%
Claude Sonnet	82%	88%	+6%

Key Insight: Memory consistently benefits stronger models. Weaker models (GPT-4o-mini) over-generalize from retrieved examples and show no gain this is a model capability threshold finding, not a system bug. Results held consistent when scaled from 100 to 1000 samples, confirming gains are not a statistical artifact.

Experiment 3 Seeded Memory Experiment (GPT-4o, 1000 Test Cases)

Setup: Pre-loaded the memory system with annotated ground truth examples before evaluation to test cold-start performance.

Memory Configuration	Accuracy
No memory (stateless)	78%

Memory Configuration	Accuracy
Organic memory (learned during eval)	85%
Seeded with 50 ground truth examples	89%
Seeded with 100 ground truth examples	91%

Key Insight: Pre-loading memory with annotated ground truths significantly improves cold-start performance. This validates the seeded memory approach as a strong initialization strategy, enabling the system to perform well from the very first evaluation.

Experiment 4 Retrieval Ablation Study (GPT-4o)

Setup: Tested each retrieval stage in isolation and in combination to understand the individual contribution of each stage.

Configuration	Accuracy
Stage 1 only (similar policies)	~80%
Stage 1 + Stage 2 (+ contrasting examples)	~83%
Stage 1 + Stage 2 + Stage 3 (full system)	85%

Key Insight: Each stage contributes incrementally. Contrasting examples (Stage 2) provide the largest single gain, while semantic patterns (Stage 3) add meaningful but smaller improvements. This justifies the 3-stage design.

Experiment 5 Memory Size Scaling (Seeded, GPT-4o)

Setup: Extended the seeded memory experiment to larger seed sizes to identify the point of diminishing returns.

Seeded Examples	Accuracy
0 (stateless)	78%
50	89%
100	91%
200	~93%
500	~94%
1000	~94–95%

Key Insight: Significant gains up to ~200 examples, with diminishing returns beyond that. This identifies a practical sweet spot for memory initialization you

do not need thousands of examples to get strong performance.

Experiment 6 Cross-Domain Generalization (EvalsBench → DevAI)

Setup: Trained memory on EvalsBench and tested on the DevAI dataset to evaluate whether memory gains transfer across domains.

Setting	Accuracy
Stateless	~75–78%
With Memory (EvalsBench-trained)	~81–83%

Key Insight: Memory gains transfer across domains, with a modest drop compared to in-domain results. This confirms the system generalizes rather than overfitting to a specific dataset an important finding for real-world applicability.

Experiment 7 Cold Start Learning Curve (GPT-4o, Organic Memory)

Setup: Tracked accuracy sample-by-sample as memory grows organically from 0 examples, simulating a real deployment scenario.

Memory Size	Accuracy
0 examples	78%
50 examples	~81%
100 examples	~83%
200 examples	~84%
500+ examples	85%

Key Insight: The system improves rapidly in the early stages and stabilizes as memory matures. This learning curve demonstrates that MAJ becomes useful quickly and continues to improve over time.

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4. Summary of Findings

Finding	Evidence
Memory consistently improves stronger models	+6–10% across GPT-4o and Claude Sonnet

Finding	Evidence
Weaker models cannot leverage memory	GPT-4o-mini shows 0% gain
Seeded memory dramatically improves cold-start	78% → 91% with 100 ground truths
All 3 retrieval stages contribute incrementally	Ablation study confirms each stage adds value
Diminishing returns after ~200 seeded examples	Scaling experiment identifies sweet spot
Memory gains transfer across domains	Cross-domain experiment confirms generalizability
System learns quickly in organic deployment	Cold-start curve shows rapid early improvement

6. Plan Going Forward

Priority	Task
High	Finalize thesis paper structure around core MAJ system
High	Write up experiments 1–7 as formal results sections
Medium	Explore submitting findings to a workshop or conference